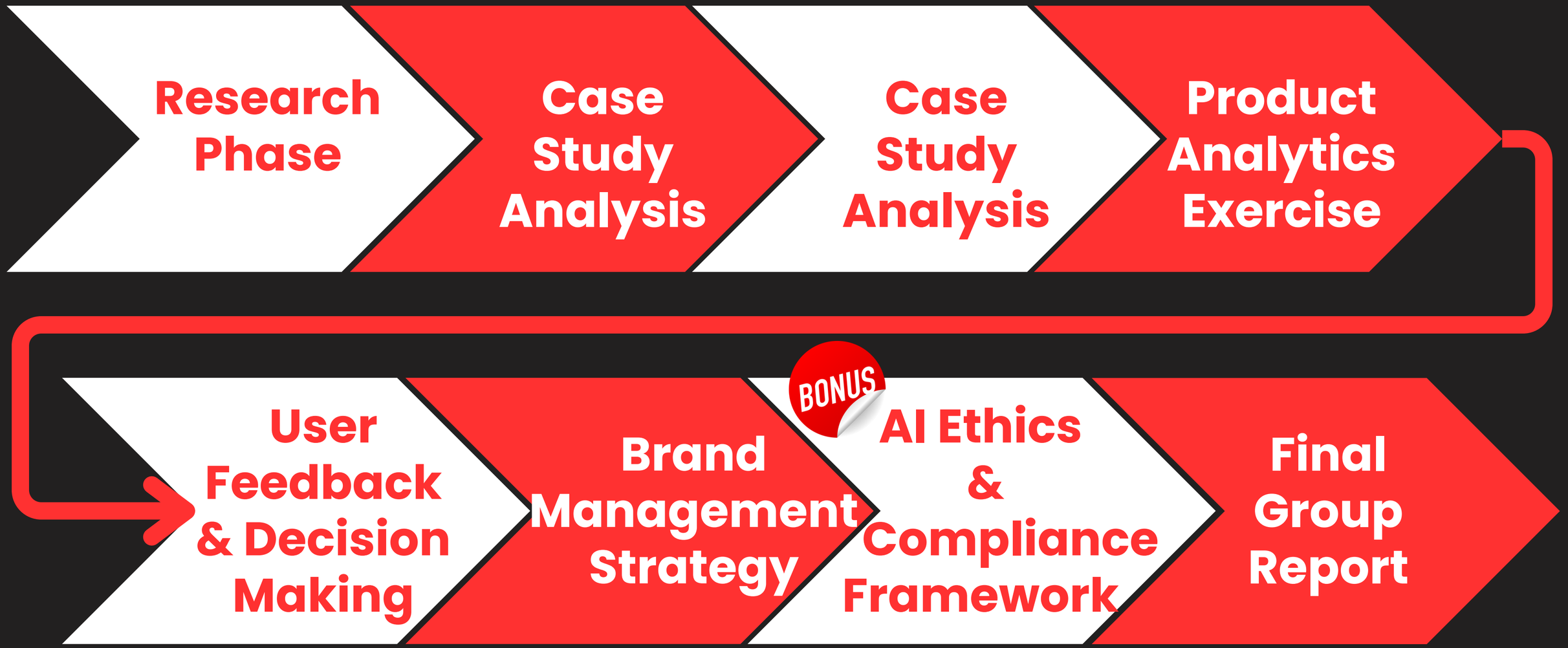
BANK OF AMERICA FRAUD DETECTION

BY: GREG FYFE, ALI SHAKIR, ELIZABETH ENGEL, HUNTER JABLONSKI, JOSEPH SRIRAMAN, KENDRA BUCCI, LEXIE BRADFORD, & TYLER EVANGELISTA

THE ASSIGNMENT

Today, we embark on a journey to discover how Bank of America harnesses the power of AI in fraud detection, a pivotal chapter in our exploration of artificial intelligence's role in product analytics, decision-making, and brand management.

OUR PROCESS (AGENDA)



RESEARCH PHASE

AI CAPABILTIES

- Data Analysis: Utilizes ML to scan extensive customer transaction data.
- Profile Creation: Builds detailed customer profiles based on spending behaviors and patterns.
- Fraud Flagging: Flags transactions deviating from the norm as potentially fraudulent.
- Evolving Fraud Profiles: Continuously updates fraud profiles for more efficient detection.
- Learning and Adapting: AI system improves over time in distinguishing between legitimate and fraudulent transactions.

- Transaction Security: Enhances overall security of the bank's transaction processing system.
- Reduced Investigation Time: AI and ML technologies have significantly decreased fraud investigation durations.
- Improved Detection Accuracy: Increases accuracy of fraud detection.
- Fraud Scores Assignment: Assigns fraud scores to transactions based on various parameters, including amount, frequency, and location.
- Real-time Risk Assessment: Uses AI for real-time fraud assessment, securing banking experiences.

POTENTIAL APPLICATIONS IN BRAND AND PRODUCT MANAGEMENT

AI in Market Trend Prediction

- Utilizes ML for accurate cash flow predictions with tools like CashPro Forecasting.
- Automates and improves accuracy in financial forecasting and market trends analysis

Fraud Prevention in Online Retail

- AI monitors customer transactions to detect and prevent online retail fraud.
- Identifies unusual behavior patterns to flag suspicious transactions.

Customer Review Analysis

- Analyzes large volumes of customer feedback for insights.
- Helps in understanding customer preferences and improving products and services.

ASSUMPTIONS

- Assumes that AI's capability to monitor and analyze transaction patterns is crucial in detecting and preventing fraud in online retail.
- Understands the importance of AI in safeguarding online transactions and enhancing customer trust.

WHAT I WOUD LIKE TO SEE

- Wishes to see insights on user experience improvements directly attributable to AI interventions.
- Hopes to see discussions on the limitations and challenges of implementing AI in these areas.

CASE STUDY ANALYSIS

ROLE OF ARTIFICIAL INTELLIGENCE IN FINANCIAL FRAUD DETECTION; ACADEMY OF MARKETING STUDIES JOURNAL

- Integration with Legacy Systems: Integrating AI solutions with existing banking systems without operational disruptions.
- Regulatory Adherence: Aligning AI implementations with strict financial sector regulations.
- High Initial Investment: Substantial initial costs, particularly challenging for smaller entities.
- Customer Trust: Building trust in AI systems amidst concerns over reduced human interaction and potential errors.
- Evolving Fraud Tactics: Necessity for AI solutions to continuously update against sophisticated and changing fraudulent activities.

- Integration with Legacy Systems: Implement gradual AI integration and partner with tech firms for seamless integration.
- Regulatory Adherence: Update AI algorithms regularly for compliance and stay proactive with financial regulations.
- High Initial Investment: Form partnerships with AI tech firms for cost-effective solutions.
- Customer Trust: Educate customers on AI benefits and maintain transparency in operations.
- Evolving Fraud Tactics: Continually adapt and update AI algorithms against advanced fraud techniques.

IMPACT OF AI IN THE CASE STUDY

Brand's Performance

- Enhanced fraud detection rates and reduced false positives.
- Streamlined fraud detection processes lead to cost savings.
- Improved risk prediction and management in financial fraud.

Customer Perception

- Increased trust and satisfaction due to effective consumer protection by AI.
- Institutions using AI perceived as innovative, enhancing market competitiveness.

ASSUMPTIONS

- Effective solutions include robust security measures, gradual integration, and customer education.
- Improved AI systems can lead to increased customer trust and satisfaction

WHAT I WOULD LIKE TO SEE

- Statistical evidence showing the impact of AI on performance metrics and customer perception.
- Comparison between AI and traditional methods in fraud detection.

PRODUCT ANALYTICS EXERCISE

In our project, we're using a simulated dataset provided by our instructor to analyze product performance with AI tools and identify key performance indicators to forecast future product trends.

EXERCISE INSTRUCTIONS

HANDED → OUT DATA

Monthly View	Daily Active Users	Retention Rates	Highest Acquisition Source	Avg. Rev Metrics per User/mnth	Average User Reviews (1-5)	% of IOS Device	Feature Used Most	Feature Used Least
12/22	1562	90.00%	social media	$102.00	4.6	60.00%	A	D
01/23	2100	87.00%	organic search	$78.00	4.6	59.00%	A	D
02/23	804	91.00%	organic search	$81.00	4.3	58.00%	B	A
03/23	1762	77.00%	organic search	$115.00	4.8	58.00%	B	A
04/23	981	81.00%	organic search	$131.00	4.8	60.00%	A	B
05/23	1004	92.00%	social media	$96.00	4.1	71.00%	A	B
06/23	2189	92.00%	ads	$71.00	4.1	71.00%	C	D
07/23	4012	89.00%	ads	$65.00	4.3	73.00%	C	D
08/23	3841	98.00%	ads	$89.00	4.4	74.00%	C	D
09/23	675	75.00%	organic search	$131.00	4.4	72.00%	A	B
10/23	896	50.00%	organic search	$142.00	4.7	70.00%	C	A

OUR FINDINGS

Daily Active Users and Revenue Analysis:

- July 2023: DAU - 4,012 | Revenue - $65
- August 2023: DAU - 3,841 | Revenue - $89
- June 2023: DAU - 2,189 | Revenue - $71
- Usage Trends: Highest use of Feature C, least use of Feature D across all months.

Feature Usage Analysis on iOS Devices:

- iOS Device Usage Below 70%:
  - Most Used Features: C or A
  - Least Used Features: D and B
- iOS Device Usage Above 70%:
  - Most Used Features: A or B
  - Least Used Feature: D

Social Media Impact on User Acquisition:

- December 2022 Insights:
  - Highest user acquisition from social media.
  - Achieved 1,562 users with a 90% retention rate and generated $102 revenue.

WHAT I WOULD LIKE TO SEE

Social Media Impact on User Acquisition:

- Curiosity about changes in social media marketing strategies.
- Wondering why social media marketing was reduced or altered after its success.
- Interest in seeing data if social media strategies reverted to December 2022's approach.
- Keen to compare Monthly Active Users (MAU) and performance in other months to December 2022, given social media's effectiveness.

Feature Usage Analysis on iOS Devices:

- Questioning feature compatibility on different iOS and Android devices.
- Interest in comparing feature usage data for Android devices.
- Desire to analyze features used on iOS vs. Android.

USER FEEDBACK AND DECISION MAKING

- **Implementation of AI & Adaptive Learning**

  The new AI system at Bank of America integrates state-of-the-art fraud detection with existing AI technologies. This system is designed to process and analyze extensive user feedback data, utilizing machine learning for continual improvement. It features advanced adaptive learning mechanisms, allowing it to constantly update its algorithms based on emerging fraudulent activities. This ensures the AI system remains proactive and ahead in detecting and preventing new fraud trends.

HYPOTHETICALLY BANK OF AMERICA IS SET TO LAUNCH AN ADVANCED FRAUD DETECTION AI AIMED AT BOLSTERING SECURITY AND ANALYZING USER FEEDBACK TO PREVENT FRAUD AND ENHANCE THE CUSTOMER BANKING EXPERIENCE.

- **User Feedback Analysis-**
  Customers are strongly encouraged to provide feedback through different channels including online surveys, mobile application reviews, and customer service interactions. The AI system processed both structured and unstructured data, extracting valuable insights from text-based comments, ratings, and reviews. The feedback included information about transactions, account access, customer service interactions, and overall satisfaction.

DECISION MAKING FRAMEWORK

User Feedback Collection

Collect user feedback through different channels and Integrate Bank of America's Fraud Detection AI to analyze not only general satisfaction but also identify any concerns related to security or suspicious activities.

Impact Assessment

Integrate machine learning algorithims that assess the potential impact of a concenr on the user expierence, product fucntion, and overall customer satisfactions. Also consider feedback for potential positive impact for fixing certain issues.

Root Cause Analysis

Use AI technology to identify underlying causes of issues that are recurring by analyzing data from the past and other patterns. Determine whether concerns are secluded or whether there is a broader problem that shows a pattern.

ASSUMPTIONS

- Believes continuous AI updates are crucial for staying ahead of fraud.

WHAT I WOULD LIKE TO SEE

- Interested in real-world examples where similar AI implementation improved user experience and fraud detection.

BRAND MANAGEMENT STRATEGY

Potential Brand Strategy Utilizing AI

- As product managers of BOA's fraud detection AI services, an improved brand strategy would be to leverage AI algorithms across social media. Social media monitoring AI features would provide additional insights into the target market and their behaviors. The greater data collected through social media can serve as a way to establish a more accurate fraud detection system.

AI in Brand Analysis

- BOA primarily utilizes X, Facebook, Linkedin, and YouTube as its media presence. Employing AI across these platforms can be in the form of social media listening, where AI can track the comments and sentiments of consumers. The keywords tracked and related to BOA, its products, and services can allow the brand to understand the public's perception of the brand and its AI application in fraud detection.

AI in Content Creation

- AI-driven content creation streamlines the process, enhancing quality and relevance for Bank of America's audience.
- Past content analysis informs future content strategy in advertisements, social media, and new services.

Key Metrics

- Likes/Reactions for positive engagement.
- Comments for consumer sentiment.
- Shares/Retweets to gauge promotional willingness.
- Reach/Impressions of promotional content.
- Customer feedback and reviews.
- Link clicks to measure application interest.

Predictive Analytics

- Cross-referencing BOA's fraud detection data with social media AI.
- Tracks consumer behaviors like spending habits and location of transactions.
- Integrates online behavior data from social platforms for deeper insights.
- Example: AI flags unusual spending but confirms legitimacy through social media activity.

ASSUMPTIONS

- Believes in the potential of AI for social media listening to understand public perception and improve brand image.
- Assumes that leveraging AI on social media platforms can enhance brand strategy and fraud detection accuracy

WHAT I WOULD LIKE TO SEE

- Curiosity about detailed insights into consumer behavior derived from AI analysis on social media.
- Wishes to see a comparison of outcomes between AI-driven social media and traditional brand management strategies.

EXTRA

AI ETHICS AND COMPLIANCE FRAMEWORK

ETHICAL CONSIDERATIONS AND REGULATORY COMPLIANCE

Ethical Considerations

- Assessing AI for privacy concerns, ensuring that customer data is used responsibly and securely, with a strong focus on consent and data rights.
- Examining algorithmic bias, particularly in fraud detection, to prevent unfair profiling or discrimination against certain customer groups.
- Addressing customer trust, which involves transparent communication about how AI is used and its impacts on customer services and privacy.

Regulatory Compliance

- Analyzing compliance with regulations such as GDPR and CCPA, focusing on how the AI system adheres to data protection and privacy standards.
- Outlining strategies to keep the AI system aligned with these laws, including data processing transparency, customer consent mechanisms, and rights to data access and erasure.
- Preparing for potential future regulatory changes by staying informed and agile in adapting AI systems to new legal requirements.

DATA PRACTICES AND STAKEHOLDER COMMUNICATION

Data Privacy and Security

Collect user feedback through different channels and Integrate Bank of America's Fraud Detection AI to analyze not only general satisfaction but also identify any concerns related to security or suspicious activities.

Transparency and Accountability

Integrate machine learning algorithims that assess the potential impact of a concenr on the user expierence, product fucntion, and overall customer satisfactions. Also consider feedback for potential positive impact for fixing certain issues.

Monitoring, Auditing, and Communication

Use AI technology to identify underlying causes of issues that are recurring by analyzing data from the past and other patterns. Determine whether concerns are secluded or whether there is a broader problem that shows a pattern.

ASSUMPTIONS

- Recognizes the critical role of ethics and compliance in AI applications within banking, especially for fraud detection.
- Acknowledges the challenges in maintaining data privacy and security in AI systems.

WHAT I WOULD LIKE TO SEE

- Curious about the direct impact of ethical and regulatory compliance on AI system effectiveness.
- Wishes to see an evaluation of the effectiveness of proposed frameworks in real-world scenarios.

FINAL GROUP REPORT

Overview:

- Focused on integrating AI into various aspects of banking operations, especially in fraud detection.
- Assessed AI's role in enhancing product management and customer experience.

Key Findings:

- AI's efficiency in analyzing and learning from transactional data and user feedback.
- Successful implementation of AI to predict fraud and analyze market trends.
- Challenges and solutions in AI integration, such as data privacy and adapting to regulations.

Strategic Insights:

- AI's impact on enhancing operational efficiency, customer trust, and brand reputation.
- Need for continuous innovation and adaptation to evolving fraud tactics.

Conclusion:

- AI proves to be an indispensable tool in banking for fraud detection and operational efficiency.
- Ongoing research is essential to fully leverage AI's potential in different banking segments.

THANK YOU

RESOOURCES